

Many printers are capable of printing multiple objects allowing users to carry out simultaneous manufacturing tasks for maximum efficiency.

The printing process

As you can see, the generic workflow of a 3D printing process is divided into a series of stages. The most important of which is obviously the printing stage where the actual magic of 3D printing happens. The system that governs how 3D printers take a digitally created computer-aided design or a scanned object file, and convert it into a layer-based printing protocol is worth understanding so that users can maximise the utility of their workflow.

The 3D printer has been created to interpret the STL file format of an object's design and deconstruct it into layered segments. Imagine a loaf of bread that has been sliced into hundreds and thousands of thin layers. The ability to craft layer upon layer of this bread in perfect alignment and fusing them into one single object gives you the loaf of bread that those slices were a part of originally. Similarly, a 3D printer digitally dismantles an object into numerous layers and prints them in perfect alignment.

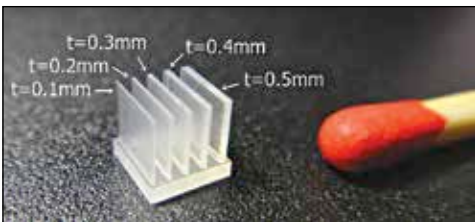
Depending on the type of printer and the object being printed, the material used can be liquid, powder, paper, metal or even food based. The printer uses the cross-sections of the object to create the final item. Since almost every object can be physically broken down into layers, the reverse process of joining together the layers is made simple by the printer. The printer

takes the virtual cross-sections from the computer-aided design model and replicates its geometric shape in layered stages.

The degree of complexity and finesse with which a printer can create objects is based



The slice-based layering of objects is similar to the deconstruction of a loaf of bread



3D printers are capable of extremely small layers providing detailed resolutions